1)

$I_n = \int x^n \mathrm{e}^x \, \mathrm{d}x$

Use Integration by Parts:

$$u = x^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1} \qquad v = \mathrm{e}^x$$

$\therefore \int x^n \mathrm{e}^x \, \mathrm{d}x = x^n \mathrm{e}^x - n \int x^{n-1} \mathrm{e}^x \, \mathrm{d}x$

$= x^n \mathrm{e}^x - nI_{n-1}$ as req.

2) a)

$I_0 = \int \mathrm{e}^x \, \mathrm{d}x = \mathrm{e}^x + c$

$I_2 = x^2 \mathrm{e}^x - 2I_1$

$= x^2 \mathrm{e}^x - 2(x\mathrm{e}^x - I_0) = x^2 \mathrm{e}^x - 2x\mathrm{e}^x + 2I_0$

$= x^2 \mathrm{e}^x - 2x\mathrm{e}^x + 2\mathrm{e}^x + c$

$= \mathrm{e}^x(x^2 - 2x + 2) + c$

2) b)

$I_0 = \int \mathrm{e}^x \, \mathrm{d}x = \mathrm{e}^x + c$

$I_3 = x^3 \mathrm{e}^x - 3I_2$

$= x^3 \mathrm{e}^x - 3(x^2 \mathrm{e}^x - 2I_1) = x^3 \mathrm{e}^x - 3x^2 \mathrm{e}^x + 6I_1$

$= x^3 \mathrm{e}^x - 3x^2 \mathrm{e}^x + 6(x\mathrm{e}^x - I_0) = x^3 \mathrm{e}^x - 3x^2 \mathrm{e}^x + 6x\mathrm{e}^x - 6I_0$

$= x^3 \mathrm{e}^x - 3x^2 \mathrm{e}^x + 6x\mathrm{e}^x - 6\mathrm{e}^x + c$

$= \mathrm{e}^x(x^3 - 3x^2 + 6x - 6) + c$

2) c)

$I_0 = \int \mathrm{e}^x \, \mathrm{d}x = \mathrm{e}^x + c$

$I_4 = x^4 \mathrm{e}^x - 4I_3$

$= x^4 \mathrm{e}^x - 4(x^3 \mathrm{e}^x - 3I_2) = x^4 \mathrm{e}^x - 4x^3 \mathrm{e}^x + 12I_2$

$= x^4 \mathrm{e}^x - 4x^3 \mathrm{e}^x + 12(x^2 \mathrm{e}^x - 2I_1) = x^4 \mathrm{e}^x - 4x^3 \mathrm{e}^x + 12x^2 \mathrm{e}^x - 24I_1$

$= x^4 \mathrm{e}^x - 4x^3 \mathrm{e}^x + 12x^2 \mathrm{e}^x - 24(x\mathrm{e}^x - I_0) = x^4 \mathrm{e}^x - 4x^3 \mathrm{e}^x + 12x^2 \mathrm{e}^x - 24x\mathrm{e}^x + 24I_0$

$= x^4 \mathrm{e}^x - 4x^3 \mathrm{e}^x + 12x^2 \mathrm{e}^x - 24x\mathrm{e}^x + 24\mathrm{e}^x + c$

$= \mathrm{e}^x(x^4 - 4x^3 + 12x^2 - 24x + 24) + c$

3)

$I_n = \int x^n \mathrm{e}^{2x} \, \mathrm{d}x$

Use Integration by Parts:

$u = x^n$ $\qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1}$ $\qquad v = \frac{1}{2}\mathrm{e}^{2x}$

$\therefore \int x^n \mathrm{e}^{2x}\,\mathrm{d}x = \frac{1}{2}x^n\mathrm{e}^{2x} - \frac{1}{2}n\int x^{n-1}\mathrm{e}^{2x}\,\mathrm{d}x$

$= \frac{1}{2}x^n\mathrm{e}^{2x} - \frac{n}{2}I_{n-1}$ as req.

4) a)

$I_0 = \int \mathrm{e}^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{e}^{2x} + c$

$I_3 = \frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{2}I_2$

$= \frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{2}\left(\frac{1}{2}x^2\mathrm{e}^{2x} - I_1\right) = \frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{4}x^2\mathrm{e}^{2x} + \frac{3}{4}I_1$

$= \frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{4}x^2\mathrm{e}^{2x} + \frac{3}{4}\left(\frac{1}{2}x\mathrm{e}^{2x} - \frac{1}{2}I_0\right) = \frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{4}x^2\mathrm{e}^{2x} + \frac{3}{8}x\mathrm{e}^{2x} - \frac{3}{8}I_0$

$= \frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{4}x^2\mathrm{e}^{2x} + \frac{3}{8}x\mathrm{e}^{2x} - \frac{3}{16}\mathrm{e}^{2x} + c$

$= \frac{1}{8}\mathrm{e}^{2x}(4x^3 - 6x^2 + 6x - 3) + c$

4) b)

$I_0 = \int \mathrm{e}^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{e}^{2x} + c$

$I_4 = \frac{1}{2}x^4\mathrm{e}^{2x} - 2I_3$

$= \frac{1}{2}x^4\mathrm{e}^{2x} - 2\left(\frac{1}{2}x^3\mathrm{e}^{2x} - \frac{3}{2}I_2\right) = \frac{1}{2}x^4\mathrm{e}^{2x} - x^3\mathrm{e}^{2x} + 3I_2$

$= \frac{1}{2}x^4\mathrm{e}^{2x} - x^3\mathrm{e}^{2x} + 3\left(\frac{1}{2}x^2\mathrm{e}^{2x} - I_1\right) = \frac{1}{2}x^4\mathrm{e}^{2x} - x^3\mathrm{e}^{2x} + \frac{3}{2}x^2\mathrm{e}^{2x} - 3I_1$

$= \frac{1}{2}x^4\mathrm{e}^{2x} - x^3\mathrm{e}^{2x} + \frac{3}{2}x^2\mathrm{e}^{2x} - 3\left(\frac{1}{2}x\mathrm{e}^{2x} - \frac{1}{2}I_0\right)$

$= \frac{1}{2}x^4\mathrm{e}^{2x} - x^3\mathrm{e}^{2x} + \frac{3}{2}x^2\mathrm{e}^{2x} - \frac{3}{2}x\mathrm{e}^{2x} + \frac{3}{2}I_0$

$= \frac{1}{2}x^4\mathrm{e}^{2x} - x^3\mathrm{e}^{2x} + \frac{3}{2}x^2\mathrm{e}^{2x} - \frac{3}{2}x\mathrm{e}^{2x} + \frac{3}{4}\mathrm{e}^{2x} + c$

$= \frac{1}{4}\mathrm{e}^{2x}(2x^4 - 4x^3 + 6x^2 - 6x + 3) + c$

5)

$I_n = \int x\,(\ln x)^n \mathrm{d}x$

Use Integration by Parts:

$u = (\ln x)^n$ $\qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{n}{x}(\ln x)^{n-1} \qquad v = \frac{1}{2}x^2$$

$\therefore \int x\,(\ln x)^n \mathrm{d}x = \frac{1}{2}x^2(\ln x)^n - \frac{n}{2}\int x(\ln x)^{n-1}\,\mathrm{d}x$

$I_n = \frac{x^2}{2}(\ln x)^n - \frac{n}{2}I_{n-1}$ as req.

6) a)

$I_0 = \int x = \mathrm{d}x = \frac{1}{2}x^2 + c$

$I_2 = \frac{x^2}{2}(\ln x)^2 - I_1$

$= \frac{x^2}{2}(\ln x)^2 - \left(\frac{x^2}{2}(\ln x) - \frac{1}{2}I_0\right)$

$= \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2}\ln x + \frac{1}{4}x^2 + c$

$= \frac{x^2}{4}(2(\ln x)^2 - 2\ln x + 1) + c$

6) b)

$I_0 = \int x = \mathrm{d}x = \frac{1}{2}x^2 + c$

$I_3 = \frac{x^2}{2}(\ln x)^3 - \frac{3}{2}I_2$

$= \frac{x^2}{2}(\ln x)^3 - \frac{3}{2}\left(\frac{x^2}{2}(\ln x)^2 - I_1\right) = \frac{x^2}{2}(\ln x)^3 - \frac{3x^2}{4}(\ln x)^2 + \frac{3}{2}I_1$

$= \frac{x^2}{2}(\ln x)^3 - \frac{3x^2}{4}(\ln x)^2 + \frac{3}{2}\left(\frac{x^2}{2}(\ln x) - \frac{1}{2}I_0\right)$

$= \frac{x^2}{2}(\ln x)^3 - \frac{3x^2}{4}(\ln x)^2 + \frac{3x^2}{4}(\ln x) - \frac{3}{8}x^2 + c$

$= \frac{1}{8}x^2(4(\ln x)^3 - 6(\ln x)^2 + 6\ln x - 3) + c$

7)

$I_n = \int \sin^n x\,\mathrm{d}x = \int \sin x \sin^{n-1} x\,\mathrm{d}x$

Use Integration by Parts:

$$u = \sin^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = (n-1)\cos x \sin^{n-2} x \qquad v = -\cos x$$

$\therefore \int \sin x \sin^{n-1} x\,\mathrm{d}x = -\cos x \sin^{n-1} x + (n-1)\int \cos^2 x \sin^{n-2} x\,\mathrm{d}x$

$= -\cos x \sin^{n-1} x + (n-1)\int (1 - \sin^2 x)\sin^{n-2} x\,\mathrm{d}x$

$= -\cos x \sin^{n-1} x + (n-1)\int \sin^{n-2} x\,\mathrm{d}x - (n-1)\int \sin^n x\,\mathrm{d}x$

$= -\cos x \sin^{n-1} x + (n-1)I_{n-2} - (n-1)I_n$

So

$I_n = -\cos x \sin^{n-1} x + (n-1)I_{n-2} - (n-1)I_n$

$I_n + (n-1)I_n = -\cos x \sin^{n-1} x + (n-1)I_{n-2}$

$nI_n = -\cos x \sin^{n-1} x + (n-1)I_{n-2}$

$I_n = -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$ as req.

8) a)

$I_0 = \int \sin^0 x \, \mathrm{d}x = -\cos x + c$

$I_4 = -\frac{\cos x \sin^3 x}{4} + \frac{3}{4} I_2$

$= -\frac{\cos x \sin^3 x}{4} + \frac{3}{4}\left(-\frac{\cos x \sin x}{2} + \frac{1}{2} I_0\right) = -\frac{\cos x \sin^3 x}{4} - \frac{3\cos x \sin x}{8} + \frac{3}{8} I_0$

$= -\frac{\cos x \sin^3 x}{4} - \frac{3\cos x \sin x}{8} + \frac{3}{8} x + c$

8) b)

$I_1 = \int \sin^1 x \, \mathrm{d}x = -\cos x + c$

$I_5 = -\frac{\cos x \sin^4 x}{5} + \frac{4}{5} I_3$

$= -\frac{\cos x \sin^4 x}{5} + \frac{4}{5}\left(-\frac{\cos x \sin^2 x}{3} + \frac{2}{3} I_1\right) = -\frac{\cos x \sin^4 x}{5} - \frac{4\cos x \sin^2 x}{15} + \frac{8}{15} I_1$

$= -\frac{\cos x \sin^4 x}{5} - \frac{4\cos x \sin^2 x}{15} - \frac{8}{15}\cos x + c$

9)

$I_n = \int \cos^n x \, \mathrm{d}x = \int \cos x \cos^{n-1} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \cos^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -(n-1)\sin x \cos^{n-2} x \qquad v = \sin x$

$\therefore \int \cos x \cos^{n-1} x \, \mathrm{d}x = \sin x \cos^{n-1} x + (n-1)\int \sin^2 x \cos^{n-2} x \, \mathrm{d}x$

$= \sin x \cos^{n-1} x + (n-1)\int (1 - \cos^2 x)\cos^{n-2} x \, \mathrm{d}x$

$= \sin x \cos^{n-1} x + (n-1)\int \cos^{n-2} x \, \mathrm{d}x - (n-1)\int \cos^n x \, \mathrm{d}x$

$= \sin x \cos^{n-1} x + (n-1)I_{n-2} - (n-1)I_n$

So

$I_n = \sin x \cos^{n-1} x + (n-1)I_{n-2} - (n-1)I_n$

$I_n + (n-1)I_n = \sin x \cos^{n-1} x + (n-1)I_{n-2}$

$nI_n = \sin x \cos^{n-1} x + (n-1)I_{n-2}$

$I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$ as req.

10) a)

$I_0 = \int \cos^0 x \, \mathrm{d}x = x + c$

$I_4 = \frac{\sin x \cos^3 x}{4} + \frac{3}{4} I_2$

$= \frac{\sin x \cos^3 x}{4} + \frac{3}{4}\left(\frac{\sin x \cos x}{2} + \frac{1}{2} I_0\right) = \frac{\sin x \cos^3 x}{4} + \frac{3 \sin x \cos x}{8} + \frac{3}{8} I_0$

$= \frac{\sin x \cos^3 x}{4} + \frac{3 \sin x \cos x}{8} + \frac{3}{8} x + c$

10) b)

$I_1 = \int \cos^1 x \, \mathrm{d}x = \sin x + c$

$I_5 = \frac{\sin x \cos^4 x}{5} + \frac{4}{5} I_3$

$= \frac{\sin x \cos^4 x}{5} + \frac{4}{5}\left(\frac{\sin x \cos^2 x}{3} + \frac{2}{3} I_1\right) = \frac{\sin x \cos^4 x}{5} + \frac{4 \sin x \cos^2 x}{15} + \frac{8}{15} I_1$

$= \frac{\sin x \cos^4 x}{5} + \frac{4 \sin x \cos^2 x}{15} + \frac{8}{15} \sin x + c$

11) a)

$I_n = \int \sinh^n x \, \mathrm{d}x = \int \sinh x \sinh^{n-1} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \sinh^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sinh x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = (n-1) \cosh x \sinh^{n-2} x \qquad v = \cosh x$

$\therefore \int \sinh x \sinh^{n-1} x \, \mathrm{d}x = \cosh x \sinh^{n-1} x - (n-1) \int \cosh^2 x \sinh^{n-2} x \, \mathrm{d}x$

$= \cosh x \sinh^{n-1} x - (n-1) \int (1 + \sinh^2 x) \sinh^{n-2} x \, \mathrm{d}x$

$= \cosh x \sinh^{n-1} x - (n-1) \int \sinh^{n-2} x \, \mathrm{d}x - (n-1) \int \sinh^n x \, \mathrm{d}x$

$= \cosh x \sinh^{n-1} x - (n-1) I_{n-2} - (n-1) I_n$

So

$I_n = \cosh x \sinh^{n-1} x - (n-1) I_{n-2} - (n-1) I_n$

$I_n + (n-1) I_n = \cosh x \sinh^{n-1} x - (n-1) I_{n-2}$

$nI_n = \cosh x \sinh^{n-1} x - (n-1) I_{n-2}$

$I_n = \frac{\cosh x \sinh^{n-1} x}{n} - \frac{n-1}{n} I_{n-2}$ as req.

11) b)

$I_0 = \int \sinh^0 x \, \mathrm{d}x = x + c$

$I_4 = \frac{\cosh x \sinh^3 x}{4} - \frac{3}{4} I_2$

$= \frac{\cosh x \sinh^3 x}{4} - \frac{3}{4}\left(\frac{\cosh x \sinh x}{2} - \frac{1}{2} I_0\right) = \frac{\cosh x \sinh^3 x}{4} - \frac{3 \cosh x \sinh x}{8} - \frac{3}{8} I_0$

$= \frac{\cosh x \sinh^3 x}{4} - \frac{3 \cosh x \sinh x}{8} - \frac{3}{8} x + c$

12) a)

$I_n = \int \cosh^n x \, \mathrm{d}x = \int \cosh x \cosh^{n-1} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \cosh^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \cosh x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = (n-1) \sinh x \cosh^{n-2} x \qquad v = \sinh x$

$\therefore \int \cosh x \cosh^{n-1} x \, \mathrm{d}x = \sinh x \cosh^{n-1} x - (n-1) \int \sinh^2 x \cosh^{n-2} x \, \mathrm{d}x$

$= \sinh x \cosh^{n-1} x - (n-1) \int (\cosh^2 x - 1) \cosh^{n-2} x \, \mathrm{d}x$

$= \sinh x \cosh^{n-1} x - (n-1) \int \cosh^n x \, \mathrm{d}x + (n-1) \int \cosh^{n-2} x \, \mathrm{d}x$

$= \sinh x \cosh^{n-1} x - (n-1) I_n + (n-1) I_{n-2}$

So

$I_n = \sinh x \cosh^{n-1} x - (n-1) I_n + (n-1) I_{n-2}$

$I_n + (n-1) I_n = \sinh x \cosh^{n-1} x + (n-1) I_{n-2}$

$n I_n = \sinh x \cosh^{n-1} x + (n-1) I_{n-2}$

$I_n = \frac{\sinh x \cosh^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$ as req.

12) b)

$I_0 = \int \cosh^0 x \, \mathrm{d}x = x + c$

$I_6 = \frac{\sinh x \cosh^5 x}{6} + \frac{5}{6} I_4$

$= \frac{\sinh x \cosh^5 x}{6} + \frac{5}{6}\left(\frac{\sinh x \cosh^3 x}{4} + \frac{3}{4} I_2\right) = \frac{\sinh x \cosh^5 x}{6} + \frac{5 \sinh x \cosh^3 x}{24} + \frac{5}{8} I_2$

$= \frac{\sinh x \cosh^5 x}{6} + \frac{5 \sinh x \cosh^3 x}{24} + \frac{5}{8}\left(\frac{\sinh x \cosh x}{2} + \frac{1}{2} I_0\right)$

$= \frac{\sinh x \cosh^5 x}{6} + \frac{5 \sinh x \cosh^3 x}{24} + \frac{5 \sinh x \cosh x}{16} + \frac{5}{16} I_0$

$= \frac{\sinh x \cosh^5 x}{6} + \frac{5 \sinh x \cosh^3 x}{24} + \frac{5 \sinh x \cosh x}{16} + \frac{5}{16} x + c$

13) a)

$I_n = \int x^n \sqrt{1+x} \, \mathrm{d}x$

Use Integration by Parts:

$$u = x^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = (1+x)^{\frac{1}{2}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1} \qquad v = \frac{2}{3}(1+x)^{\frac{3}{2}}$$

$\therefore \int x^n\sqrt{1+x}\,\mathrm{d}x = \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}\int x^{n-1}(1+x)^{\frac{3}{2}}\,\mathrm{d}x$

Since $(1+x)^{\frac{3}{2}} = (1+x)(1+x)^{\frac{1}{2}}$ then

$\int x^n\sqrt{1+x}\,\mathrm{d}x = \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}\int x^{n-1}(1+x)(1+x)^{\frac{1}{2}}\,\mathrm{d}x$

$= \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}\int (x^{n-1}+x^n)(1+x)^{\frac{1}{2}}\,\mathrm{d}x$

$= \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}\int x^{n-1}(1+x)^{\frac{1}{2}}\,\mathrm{d}x - \frac{2n}{3}\int x^n(1+x)^{\frac{1}{2}}\,\mathrm{d}x$

$= \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}I_{n-1} - \frac{2n}{3}I_n$

So

$I_n = \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}I_{n-1} - \frac{2n}{3}I_n$

$I_n + \frac{2n}{3}I_n = \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}I_{n-1}$

$\left(\frac{3+2n}{3}\right)I_n = \frac{2}{3}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3}I_{n-1}$

$I_n = \frac{2}{3+2n}x^n(1+x)^{\frac{3}{2}} - \frac{2n}{3+2n}I_{n-1}$ as req.

13) b)

$I_0 = \int \sqrt{1+x}\,\mathrm{d}x = \frac{2}{3}(1+x)^{\frac{3}{2}} + c$

$I_2 = \frac{2}{7}x^2(1+x)^{\frac{3}{2}} - \frac{4}{7}I_1$

$= \frac{2}{7}x^2(1+x)^{\frac{3}{2}} - \frac{4}{7}\left(\frac{2}{5}x(1+x)^{\frac{3}{2}} - \frac{2}{5}I_0\right) = \frac{2}{7}x^2(1+x)^{\frac{3}{2}} - \frac{8}{35}x(1+x)^{\frac{3}{2}} + \frac{8}{35}I_0$

$= \frac{2}{7}x^2(1+x)^{\frac{3}{2}} - \frac{8}{35}x(1+x)^{\frac{3}{2}} + \frac{16}{105}(1+x)^{\frac{3}{2}} + c$

$= \frac{2}{105}(1+x)^{\frac{3}{2}}(15x^2 - 12x + 8) + c$

14) a)

$I_n = \int \tan^n x\,\mathrm{d}x = \int \tan^2 x \tan^{n-2} x\,\mathrm{d}x$

$= \int (\sec^2 x - 1)\tan^{n-2} x\,\mathrm{d}x$

$= \int \sec^2 x \tan^{n-2} x\,\mathrm{d}x - \int \tan^{n-2} x\,\mathrm{d}x$

$= \frac{1}{n-1}\tan^{n-1} x - I_{n-2}$ as req.

14) b)

$\int \tan x \, \mathrm{d}x = \int \frac{\sin x}{\cos x} \mathrm{d}x = -\ln|\cos x| + c = \ln\left|\frac{1}{\cos x}\right| + c = \ln|\sec x| + c$

14) c)

$I_1 = \int \tan x \, \mathrm{d}x = \ln|\sec x| + c$

$I_5 = \frac{1}{4}\tan^4 x - I_3$

$= \frac{1}{4}\tan^4 x - \left(\frac{1}{2}\tan^2 x - I_1\right) = \frac{1}{4}\tan^5 x - \frac{1}{2}\tan^2 x + I_1$

$= \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \ln|\sec x| + c$

15) a)

$I_{m,n} = \int x^m (\ln x)^n \, \mathrm{d}x$

Use Integration by Parts:

$$u = (\ln x)^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^m$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{n}{x}(\ln x)^{n-1} \qquad v = \frac{x^{m+1}}{m+1}$$

$\therefore \int x^m (\ln x)^n \, \mathrm{d}x = \frac{x^{m+1}}{m+1}(\ln x)^n - \frac{n}{m+1}\int \frac{1}{x}(\ln x)^{n-1} x^{m+1} \, \mathrm{d}x$

$= \frac{x^{m+1}}{m+1}(\ln x)^n - \frac{n}{m+1}\int x^m (\ln x)^{n-1} \, \mathrm{d}x$

$\therefore I_{m,n} = \frac{x^{m+1}}{m+1}(\ln x)^n - \frac{n}{m+1} I_{m,n-1}$

15) b)

$m = 3, n = 1$

$I_{3,0} = \int x^3 (\ln x)^0 \, \mathrm{d}x = \int x^3 \, \mathrm{d}x = \frac{1}{4}x^4$

$I_{3,1} = \frac{x^4}{4}\ln x - \frac{1}{4} I_{3,0}$

$= \frac{x^4}{4}\ln x - \frac{1}{16}x^4 + c$

16) a)

$I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} \mathrm{d}x$

Use Integration by Parts:

$$u = x^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = (1-x)^{-\frac{1}{2}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1} \qquad v = -2(1-x)^{\frac{1}{2}}$$

$\therefore \int_0^1 \frac{x^n}{\sqrt{1-x}}\,\mathrm{d}x = \left[\frac{1}{2}x^n(1-x)^{\frac{1}{2}}\right]_0^1 + 2n\int_0^1 x^{n-1}(1-x)^{\frac{1}{2}}\,\mathrm{d}x$

$= (0) - (0) + 2n\int_0^1 x^{n-1}(1-x)^{\frac{1}{2}}\,\mathrm{d}x$

But since $\sqrt{1-x} = \frac{1-x}{\sqrt{1-x}}$

$= 2n\int_0^1 x^{n-1}\frac{1-x}{\sqrt{1-x}}\,\mathrm{d}x$

$= 2n\int_0^1 \frac{x^{n-1}}{\sqrt{1-x}}\,\mathrm{d}x - 2n\int_0^1 \frac{x^n}{\sqrt{1-x}}\,\mathrm{d}x$

$= 2nI_{n-1} - 2nI_n$

So

$I_n = 2nI_{n-1} - 2nI_n$

$\Rightarrow (1+2n)I_n = 2nI_{n-1}$

$\therefore I_n = \frac{2n}{1+2n}I_{n-1}$ as req.

16) b)

$I_0 = \int_0^1 \frac{1}{\sqrt{1-x}}\,\mathrm{d}x = \left[-2(1-x)^{\frac{1}{2}}\right]_0^1 = (0) - (-2) = 2$

$I_4 = \frac{8}{9}I_3$

$= \frac{8}{9}\left(\frac{6}{7}I_2\right) = \frac{16}{21}I_2$

$= \frac{16}{21}\left(\frac{4}{5}I_1\right) = \frac{64}{105}I_1$

$= \frac{64}{105}\left(\frac{2}{3}I_0\right) = \frac{128}{315}I_0$

$= \frac{256}{315}$ as req.

17) a)

$I_n = \int_0^{\frac{\pi}{2}} x^n \sin x\,\mathrm{d}x$

Use Integration by Parts:

$$u = x^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1} \qquad v = -\cos x$$

$\therefore \int_0^{\frac{\pi}{2}} x^n \sin x\,\mathrm{d}x = [-x^n\cos x]_0^{\frac{\pi}{2}} + n\int_0^{\frac{\pi}{2}} x^{n-1}\cos x\,\mathrm{d}x$

$= (0) - (0) + \int_0^{\frac{\pi}{2}} nx^{n-1} \cos x \, \mathrm{d}x$

Use Integration by Parts:

$$u = nx^{n-1} \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = n(n-1)x^{n-2} \qquad v = \sin x$$

$\therefore \int_0^{\frac{\pi}{2}} nx^{n-1} \cos x \, \mathrm{d}x = [nx^{n-1} \sin x]_0^{\frac{\pi}{2}} - n(n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, \mathrm{d}x$

$= \left(n\left(\frac{\pi}{2}\right)^{n-1}\right) - (0) - n(n-1)I_{n-2}$

$\therefore I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$ as req.

17) b)

$I_0 = \int_0^{\frac{\pi}{2}} x^n \sin x \, \mathrm{d}x = [-\cos x]_0^{\frac{\pi}{2}} = (0) - (-1) = 1$

$I_4 = 4\left(\frac{\pi}{2}\right)^3 - 12I_2$

$= \frac{\pi^3}{2} - 12\left(2\left(\frac{\pi}{2}\right)^1 - 2I_0\right) = \frac{\pi^3}{2} - 12\pi + 24I_0$

$= \frac{\pi^3}{2} - 12\pi + 24$

$= \frac{1}{2}(48 - 24\pi + \pi^3)$ as req.

18) a)

$I_n = \int_0^{2\pi} \cos^n x \, \mathrm{d}x$

Use Integration by Parts:

$$u = \cos^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -(n-1)\sin x \cos^{n-2} x \qquad v = \sin x$$

$\therefore \int_0^{2\pi} \cos^n x \, \mathrm{d}x = [\sin x \cos^{n-1} x]_0^{2\pi} + (n-1) \int_0^{2\pi} \sin^2 x \cos^{n-2} x \, \mathrm{d}x$

$= (0) - (0) + (n-1) \int_0^{2\pi} (1 - \cos^2 x) \cos^{n-2} x \, \mathrm{d}x$

$= (n-1) \int_0^{2\pi} \cos^{n-2} x \, \mathrm{d}x - (n-1) \int_0^{2\pi} \cos^2 x \sin^{n-2} x \, \mathrm{d}x$

$= (n-1)I_{n-2} - (n-1)I_n$

So

$I_n = (n-1)I_{n-2} - (n-1)I_n$

$\Rightarrow \big(1 + (n-1)\big)I_n = (n-1)I_{n-2}$

$\therefore I_n = \frac{n-1}{n} I_{n-2}$

$I_1 = \int_0^{2\pi} \cos x \, \mathrm{d}x = [\sin x]_0^{2\pi} = (0) - (0) = 0$

So $\int_0^{2\pi} \cos^n x \, \mathrm{d}x = 0$ for odd values of n

18) b)

$I_0 = \int_0^{2\pi} 1 \, \mathrm{d}x = [x]_0^{2\pi} = (2\pi) - (0) = 2\pi$

$I_2 = \frac{1}{2} I_0 = \frac{1}{2} \times 2\pi$

$I_4 = \frac{3}{4} I_2 = \frac{3}{4} \times \frac{1}{2} \times 2\pi$

$I_6 = \frac{5}{6} I_4 = \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times 2\pi$

…

$I_{2n} = \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right) \ldots \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) 2\pi$ for even values of n

19) a)

$I_n = \int_0^{2\pi} \sin^n x \, \mathrm{d}x$

Use Integration by Parts:

$$u = \sin^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = (n-1)\cos x \sin^{n-2} x \qquad v = -\cos x$$

$\therefore \int_0^{2\pi} \sin^n x \, \mathrm{d}x = [-\cos x \sin^{n-1} x]_0^{2\pi} + (n-1)\int_0^{2\pi} \cos^2 x \sin^{n-2} x \, \mathrm{d}x$

$= (0) - (0) + (n-1)\int_0^{2\pi} (1 - \sin^2 x) \sin^{n-2} x \, \mathrm{d}x$

$= (n-1)\int_0^{2\pi} \sin^{n-2} x \, \mathrm{d}x - (n-1)\int_0^{2\pi} \sin^2 x \sin^{n-2} x \, \mathrm{d}x$

$= (n-1)I_{n-2} - (n-1)I_n$

So

$I_n = (n-1)I_{n-2} - (n-1)I_n$

$\Rightarrow \big(1 + (n-1)\big)I_n = (n-1)I_{n-2}$

$\therefore I_n = \frac{n-1}{n} I_{n-2}$

$I_1 = \int_0^{2\pi} \sin x \, \mathrm{d}x = [-\cos x]_0^{2\pi} = (-1) - (-1) = 0$

So $\int_0^{2\pi} \sin^n x \, dx = 0$ for odd values of n

19) b)

$I_0 = \int_0^{2\pi} 1 \, dx = [x]_0^{2\pi} = (2\pi) - (0) = 2\pi$

$I_2 = \frac{1}{2} I_0 = \frac{1}{2} \times 2\pi$

$I_4 = \frac{3}{4} I_2 = \frac{3}{4} \times \frac{1}{2} \times 2\pi$

$I_6 = \frac{5}{6} I_4 = \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times 2\pi$

…

$I_{2n} = \left(\frac{2n-1}{2n}\right)\left(\frac{2n-3}{2n-2}\right) \ldots \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) 2\pi$ for even values of n

20) a)

$I_n = \int_0^{\frac{\pi}{3}} \sec^n x \, dx = \int_0^{\frac{\pi}{3}} \sec^{n-2} x \sec^2 x \, dx = \int_0^{\frac{\pi}{3}} \sec^{n-2} x \sec^2 x \, dx$

Use Integration by Parts:

$$u = \sec^{n-2} x \qquad \frac{dv}{dx} = \sec^2 x$$

$$\frac{du}{dx} = (n-2) \tan x \sec^{n-3} x \qquad v = \tan x$$

$\therefore \int_0^{\frac{\pi}{3}} \sec^n x \, dx = [\tan x \sec^{n-2} x]_0^{\frac{\pi}{3}} - (n-2) \int_0^{\frac{\pi}{3}} \tan^2 x \sec^{n-2} x \, dx$

$= \sqrt{3}(2^{n-2}) - (0) - (n-2) \int_0^{\frac{\pi}{3}} (\sec^2 x - 1) \sec^{n-3} x \, dx$

$= 2^{n-2}\sqrt{3} - (n-2) \int_0^{\frac{\pi}{3}} \sec^n x \, dx + (n-2) \int_0^{\frac{\pi}{3}} \sec^{n-2} x \, dx$

$= 2^{n-2}\sqrt{3} - (n-2) I_n + (n-2) I_{n-2}$

So

$I_n = 2^{n-2}\sqrt{3} - (n-2) I_n + (n-2) I_{n-2}$

$\Rightarrow \big(1 + (n-2)\big) I_n = 2^{n-2}\sqrt{3} + (n-2) I_{n-2}$

$\therefore (n-1) I_n = 2^{n-2}\sqrt{3} + (n-2) I_{n-2}$ as req.

20) b)

$I_0 = \int_0^{\frac{\pi}{3}} 1 \, dx = [x]_0^{\frac{\pi}{3}} = \left(\frac{\pi}{3}\right) - (0) = \frac{\pi}{3}$

$I_n = \frac{2^{n-2}}{n-1}\sqrt{3} + \frac{n-2}{n-1} I_{n-2}$

$I_8 = \frac{64}{7}\sqrt{3} + \frac{6}{7} I_6$

$= \frac{64}{7}\sqrt{3} + \frac{6}{7}\left(\frac{16}{5}\sqrt{3} + \frac{4}{5} I_4\right) = \frac{416}{35}\sqrt{3} + \frac{24}{35} I_4$

$= \frac{416}{35}\sqrt{3} + \frac{24}{35}\left(\frac{4}{3}\sqrt{3} + \frac{2}{3}I_2\right) = \frac{64}{5}\sqrt{3} + \frac{16}{35}I_2$
$= \frac{64}{5}\sqrt{3} + \frac{16}{35}(\sqrt{3}) = \frac{464}{35}\sqrt{3}$

21) a)

$I_n = \int_0^{\frac{\pi}{3}} \sin^n x \, \mathrm{d}x$

Use Integration by Parts:

$$u = \sin^{n-1} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = (n-1)\cos x \sin^{n-2} x \qquad v = -\cos x$$

$\therefore \int_0^{\frac{\pi}{3}} \sin^n x \, \mathrm{d}x = \left[-\cos x \sin^{n-1} x\right]_0^{\frac{\pi}{3}} + (n-1)\int_0^{\frac{\pi}{3}} \cos^2 x \sin^{n-2} x \, \mathrm{d}x$
$= \left(-\frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)^{n-1}\right) - (0) + (n-1)\int_0^{\frac{\pi}{3}} (1 - \sin^2 x)\sin^{n-2} x \, \mathrm{d}x$
$= \left(-\frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)^{n-1}\right) + (n-1)\int_0^{\frac{\pi}{3}} \sin^{n-2} x \, \mathrm{d}x - (n-1)\int_0^{\frac{\pi}{3}} \sin^2 x \sin^{n-2} x \, \mathrm{d}x$
$= \left(-\frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)^{n-1}\right) + (n-1)I_{n-2} - (n-1)I_n$

So

$I_n = \left(-\frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)^{n-1}\right) + (n-1)I_{n-2} - (n-1)I_n$

$\Rightarrow (1 + (n-1))I_n = \frac{n-1}{n}I_{n-2} - \frac{1}{2n}\left(\frac{\sqrt{3}}{2}\right)^{n-1}$

$\therefore I_n = \frac{n-1}{n}I_{n-2} - \frac{1}{2n}\left(\frac{\sqrt{3}}{2}\right)^{n-1}$ as req.

21) b)

$I_0 = \int_0^{\frac{\pi}{3}} 1 \, \mathrm{d}x = [x]_0^{\frac{\pi}{3}} = \left(\frac{\pi}{3}\right) - (0) = \frac{\pi}{3}$

$I_6 = \frac{5}{6}I_4 - \frac{1}{12}\left(\frac{\sqrt{3}}{2}\right)^5$
$= \frac{5}{6}\left(\frac{3}{4}I_2 - \frac{1}{8}\left(\frac{\sqrt{3}}{2}\right)^3\right) - \frac{3\sqrt{3}}{128} = \frac{5}{8}I_2 - \frac{\sqrt{3}}{16}$
$= \frac{5}{8}\left(\frac{1}{2}I_0 - \frac{1}{4}\left(\frac{\sqrt{3}}{2}\right)\right) - \frac{\sqrt{3}}{16} = \frac{5}{16}I_0 - \frac{9\sqrt{3}}{64}$
$= \frac{5\pi}{48} - \frac{9\sqrt{3}}{64}$

21) c)

Since $y = \sin^6 x$ is symmetrical about $x = 0$, $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sin^6 x \, dx = \frac{5\pi}{24} - \frac{9\sqrt{3}}{32}$

22)

$I_0 = \int_0^{\frac{\pi}{2}} 1 \, dx = [x]_0^{\frac{\pi}{2}} = \left(\frac{\pi}{2}\right) - (0) = \frac{\pi}{2}$

$\int_0^{\frac{\pi}{2}} \cos^6 x \, dx = I_6 = \frac{5}{6} I_4$

$= \frac{5}{6}\left(\frac{3}{4} I_2\right) = \frac{5}{8} I_2$

$= \frac{5}{8}\left(\frac{1}{2} I_0\right) = \frac{5}{16} I_0$

$= \frac{5\pi}{32}$

Since $y = \cos^6 x$ is symmetrical about $x = 0$, $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6 x \, dx = \frac{5\pi}{16}$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6(3t) \, dt$

Consider the substitution $3t = x$:

When $t = \frac{\pi}{2}$, $t = \frac{3\pi}{2}$, when $t = -\frac{\pi}{2}$, $t = -\frac{3\pi}{2}$ and $\frac{dx}{dt} = 3$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6(3t) \, dt = \int_{-\frac{3\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{3} \cos^6 x \, dx = \frac{1}{3} \int_{-\frac{3\pi}{2}}^{\frac{3\pi}{2}} \cos^6 x \, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6 x \, dx$

$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6(3t) \, dt = \frac{5\pi}{16}$

23) a)

$I_n = \int_0^1 (1 - x^2)^n \, dx$

Use Integration by Parts:

$u = (1 - x^2)^n$ $\qquad$ $\frac{dv}{dx} = 1$

$\frac{du}{dx} = -2xn(1 - x^2)^{n-1}$ $\qquad$ $v = x$

$\therefore \int_0^1 (1 - x^2)^n \, dx = [x(1 - x^2)^n]_0^1 + 2n \int_0^1 x^2 (1 - x^2)^{n-1} \, dx$

$= (0) - (0) + 2n \int_0^1 x^2 (1 - x^2)^{n-1} \, dx$

$= 2n \int_0^1 (1 - (1 - x^2))(1 - x^2)^{n-1} \, dx$

$= 2n \int_0^1 (1 - x^2)^{n-1} \, dx - 2n \int_0^1 (1 - x^2)(1 - x^2)^{n-1} \, dx$

$= 2nI_{n-1} - 2n\int_0^1 (1-x^2)^n \,\mathrm{d}x$

$= 2nI_{n-1} - 2nI_n$

So

$I_n = 2nI_{n-1} - 2nI_n$

$\Rightarrow (1+2n)I_n = 2nI_{n-1}$

$\therefore I_n = \frac{2n}{1+2n} I_{n-1}$ as req.

23) b)

$I_0 = \int_0^1 1 \,\mathrm{d}x = [x]_0^1 = (1) - (0) = 1$

$I_6 = \frac{12}{13} I_5$

$= \frac{12}{13}\left(\frac{10}{11} I_4\right) = \frac{120}{143} I_4$

$= \frac{120}{143}\left(\frac{8}{9} I_3\right) = \frac{320}{429} I_3$

$= \frac{320}{429}\left(\frac{6}{7} I_2\right) = \frac{640}{1001} I_2$

$= \frac{640}{1001}\left(\frac{4}{5} I_1\right) = \frac{512}{1001} I_1$

$= \frac{512}{1001}\left(\frac{2}{3} I_0\right) = \frac{1024}{3003}$

Since $y = (1-x^2)^6$ is symmetrical about $x = 0$, $\int_{-1}^1 (1-x^2)^6 \,\mathrm{d}x = \frac{2048}{3003}$

24) a)

$I_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x \sin^2 x \,\mathrm{d}x$

Use Integration by Parts:

$$u = \sin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sin x \cos^{2n} x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \cos x \qquad v = -\frac{1}{2n+1}\cos^{2n+1} x$$

$\therefore \int_0^{\frac{\pi}{2}} \cos^{2n} x \sin^2 x \,\mathrm{d}x = \left[-\frac{1}{2n+1}\sin x \cos^{2n+1} x\right]_0^{\frac{\pi}{2}} + \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos x \cos^{2n+1} x \,\mathrm{d}x$

$= (0) - (0) + \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n+2} x \,\mathrm{d}x$

$= \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \cos^2 x \,\mathrm{d}x$

$= \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x (1 - \sin^2 x)\mathrm{d}x$

$= \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \,\mathrm{d}x - \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \sin^2 x \,\mathrm{d}x$

$= \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \, \mathrm{d}x - \frac{1}{2n+1} I_n$

So

$I_n = \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \, \mathrm{d}x - \frac{1}{2n+1} I_n$

$\Rightarrow \left(1 + \frac{1}{2n+1}\right) I_n = \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \, \mathrm{d}x$

$\frac{2n+2}{2n+1} I_n = \frac{1}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n} x \, \mathrm{d}x$

$I_n = \frac{1}{2n+2}\int_0^{\frac{\pi}{2}} \cos^{2n} x \, \mathrm{d}x$

Since $J_n = \int_0^{\frac{\pi}{2}} \cos^{n} x \, \mathrm{d}x$ then $I_n = \frac{1}{2n+2} J_{2n}$

$J_{2n} = \frac{2n-1}{2n} J_{2n-2}$

So

$I_n = \frac{1}{2n+2} J_{2n} = \frac{2n-1}{2n(2n+2)} J_{2n-2}$

Since $I_{n-1} = \frac{1}{2(n-1)+2} J_{2(n-1)} = \frac{1}{2n} J_{2n-2}$ then

$I_n = \frac{2n-1}{2n(2n+2)} \times 2n I_{n-1} = \frac{2n-1}{2n+2} I_{n-1}$

$\therefore I_n = \frac{2n-1}{2n+2} I_{n-1}$ as req.

24) b)

$I_0 = \int_0^{\frac{\pi}{2}} \sin^2 x \, \mathrm{d}x = \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) \mathrm{d}x = \left[\frac{1}{2}x - \frac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}} = \left(\frac{\pi}{4}\right) - (0) = \frac{\pi}{4}$

$I_6 = \frac{11}{14} I_5$

$= \frac{11}{14}\left(\frac{9}{12} I_4\right) = \frac{33}{56} I_4$

$= \frac{33}{56}\left(\frac{7}{10} I_3\right) = \frac{33}{80} I_3$

$= \frac{33}{80}\left(\frac{5}{8} I_2\right) = \frac{33}{128} I_2$

$= \frac{33}{128}\left(\frac{3}{6} I_1\right) = \frac{33}{256} I_1$

$= \frac{55}{256}\left(\frac{1}{4} I_0\right) = \frac{33}{1024} I_0$

$= \frac{33\pi}{4096}$ as req.

24) c)

Since $y = \cos^{12} x \sin^2 x$ is symmetrical about $x = 0$, $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^{12} x \sin^2 x \, \mathrm{d}x = \frac{33\pi}{2048}$

25) a)

$I_n = \int_0^{\ln 3} \tanh^n x \, \mathrm{d}x = \int_0^{\ln 3} \tanh^{n-2} x \tanh^2 x \, \mathrm{d}x$

$= \int_0^{\ln 3} \tanh^{n-2} x \, (1 - \operatorname{sech}^2 x) \mathrm{d}x$

$= \int_0^{\ln 3} \tanh^{n-2} x \, \mathrm{d}x - \int_0^{\ln 3} \tanh^{n-2} x \operatorname{sech}^2 x \, \mathrm{d}x$

Use Integration by Parts:

$$u = \tanh^{n-2} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \operatorname{sech}^2 x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = (n-2) \operatorname{sech}^2 x \tanh^{n-3} x \qquad v = \tanh x$$

$\therefore \int_0^{\ln 3} \tanh^{n-2} x \operatorname{sech}^2 x \, \mathrm{d}x = [\tanh x \tanh^{n-2} x]_0^{\ln 3} - (n-2) \int_0^{\ln 3} \operatorname{sech}^2 x \tanh^{n-2} x \, \mathrm{d}x$

$= \frac{4}{5} \times \left(\frac{4}{5}\right)^{n-2} - (0) - (n-2) \int_0^{\ln 3} (1 - \tanh^2 x) \tanh^{n-2} x \, \mathrm{d}x$

$= \left(\frac{4}{5}\right)^{n-1} - (n-2) \int_0^{\ln 3} \tanh^{n-2} x \, \mathrm{d}x + (n-2) \int_0^{\ln 3} \tanh^n x \, \mathrm{d}x$

$= \left(\frac{4}{5}\right)^{n-1} - (n-2) I_{n-2} + (n-2) I_n$

So

$I_n = \int_0^{\ln 3} \tanh^{n-2} x \, \mathrm{d}x - \left(\left(\frac{4}{5}\right)^{n-1} - (n-2) I_{n-2} - (n-2) I_n \right)$

$= I_{n-2} - \left(\frac{4}{5}\right)^{n-1} - (n-2) I_{n-2} + (n-2) I_n$

$\Rightarrow \big(1 + (n-2)\big) I_n = -\left(\frac{4}{5}\right)^{n-1} + (n-1) I_{n-2}$

$\therefore I_n = -\frac{1}{n-1} \left(\frac{4}{5}\right)^{n-1} + I_{n-2}$ as req.

25) b)

$I_0 = \int_0^{\ln 3} 1 \, \mathrm{d}x = [x]_0^{\ln 3} = (\ln 3) - (0) = \ln 3$

$I_4 = -\frac{1}{3} \left(\frac{4}{5}\right)^3 + I_2$

$= -\frac{64}{375} + \left(-\left(\frac{4}{5}\right)^1 + I_0 \right)$

$= -\frac{64}{375} - \frac{4}{5} + \ln 3 = -\frac{364}{375} + \ln 3$

26) a)

$I_n = \int_0^1 x^n \mathrm{e}^{ax} \, \mathrm{d}x$

Use Integration by Parts:

$$u = x^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{ax}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1} \qquad v = \frac{1}{a}\mathrm{e}^{ax}$$

$\therefore \int_0^1 x^n \mathrm{e}^{ax}\,\mathrm{d}x = \left[\frac{1}{a}x^n \mathrm{e}^{ax}\right]_0^1 - \frac{1}{a}n\int_0^1 x^{n-1}\mathrm{e}^{ax}\,\mathrm{d}x$

$= \left(\frac{1}{a}\mathrm{e}^a\right) - (0) - \frac{1}{a}nI_{n-1}$

$\therefore I_n = \frac{\mathrm{e}^a}{a} - \frac{n}{a}I_{n-1}$ as req.

26) b)

$I_0 = \int_0^1 \mathrm{e}^{\frac{1}{4}x}\,\mathrm{d}x = \left[4\mathrm{e}^{\frac{1}{4}x}\right]_0^1 = 4\mathrm{e}^{\frac{1}{4}} - 4$

And $I_n = 4\mathrm{e}^{\frac{1}{4}} - 4nI_{n-1}$

$I_2 = 4\mathrm{e}^{\frac{1}{4}} - 8I_1$

$= 4\mathrm{e}^{\frac{1}{4}} - 8\left(4\mathrm{e}^{\frac{1}{4}} - 4I_0\right) = -28\mathrm{e}^{\frac{1}{4}} + 32I_0 = -28\mathrm{e}^{\frac{1}{4}} + 32\left(4\mathrm{e}^{\frac{1}{4}} - 4\right) = 100\mathrm{e}^{\frac{1}{4}} - 128$

27) a)

$I_n = \int_0^{\sqrt{a}}(a - x^2)^n\,\mathrm{d}x$

Use Integration by Parts:

$$u = (a - x^2)^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -2nx(a - x^2)^{n-1} \qquad v = x$$

$\therefore \int_0^{\sqrt{a}}(a - x^2)^n\,\mathrm{d}x = [x(a - x^2)^n]_0^{\sqrt{a}} + 2n\int_0^{\sqrt{a}} x^2(a - x^2)^{n-1}\,\mathrm{d}x$

$= (0) - (0) + 2n\int_0^{\sqrt{a}}\big(a - (a - x^2)\big)(a - x^2)^{n-1}\,\mathrm{d}x$

$= 2na\int_0^{\sqrt{a}}(a - x^2)^{n-1}\,\mathrm{d}x - 2n\int_0^{\sqrt{a}}(a - x^2)^n\,\mathrm{d}x$

$= 2naI_{n-1} - 2nI_n$

So

$I_n = 2naI_{n-1} - 2nI_n$

$\Rightarrow (1 + 2n)I_n = 2naI_{n-1}$

$\therefore I_n = \frac{2na}{1+2n}I_{n-1}$ as req.

27) b)

$I_0 = \int_0^4 1\,\mathrm{d}x = [x]_0^4 = 4$

And $I_n = \frac{32n}{1+2n} I_{n-1}$

$I_3 = \frac{96}{7} I_2 = \frac{96}{7}\left(\frac{64}{5} I_1\right) = \frac{6144}{35} I_1 = \frac{6144}{35}\left(\frac{32}{3} I_0\right) = \frac{65536}{35} \times 4 = \frac{262144}{35}$

28) a)

Let $A = (n-1)x$ and $B = x$ so $A + B = nx$ and $A - B = (n-2)x$

$\cos(nx) + \cos\big((n-2)x\big) = 2\cos\big((n-1)x\big)\cos(x)$

$\Rightarrow \cos(nx) = 2\cos\big((n-1)x\big)\cos(x) - \cos\big((n-2)x\big)$

$\Rightarrow \frac{\cos(nx)}{\cos x} = 2\cos\big((n-1)x\big) - \frac{\cos\big((n-2)x\big)}{\cos x}$ as req.

28) b)

$I_n = \int_0^{\pi} \frac{\cos nx}{\cos x}\,\mathrm{d}x$

$= \int_0^{\pi} \left(2\cos\big((n-1)x\big) - \frac{\cos\big((n-2)x\big)}{\cos x}\right)\mathrm{d}x$

$= \int_0^{\pi} 2\cos\big((n-1)x\big)\,\mathrm{d}x - \int_0^{\pi} \frac{\cos\big((n-2)x\big)}{\cos x}\,\mathrm{d}x$

$= \left[\frac{2}{n-1}\sin\big((n-1)x\big)\right]_0^{\pi} - I_{n-2}$

$= (0) - (0) - I_{n-2}$

$\therefore I_n = -I_{n-2}$ as req.

28) c)

$I_1 = \int_0^{\pi} \frac{\cos x}{\cos x}\,\mathrm{d}x = \int_0^{\pi} 1\,\mathrm{d}x = [x]_0^{\pi} = (\pi) - (0) = \pi$

$I_7 = -I_5 = I_3 = -I_1 = -\pi$

28) d)

Since $I_n = -I_{n-2}$ every step, the sign changes.

$\frac{53-1}{2} = 26$ steps

$I_{53} = (-1)^{26} I_1 = \pi$

$\therefore \int_0^{\pi} \frac{\cos 53x}{\cos x}\,\mathrm{d}x = \pi$

29) a)

$I_n = \int_0^a x^n (a-x)^m\,\mathrm{d}x$

Use Integration by Parts:

$u = x^n$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = (a-x)^m$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = nx^{n-1} \qquad v = -\frac{1}{m+1}(a-x)^{m+1}$$

$\therefore \int_0^k x^n(a-x)^m \,\mathrm{d}x = \left[-\frac{1}{m+1}x^n(a-x)^{m+1}\right]_0^k + \frac{n}{m+1}\int_0^k x^{n-1}(a-x)^{m+1}\,\mathrm{d}x$

$= \left(-\frac{1}{m+1}k^n(a-k)^{m+1}\right) - (0) + \frac{n}{m+1}I_{m+1,n-1}$

$\therefore I_{m,n} = -\frac{1}{m+1}k^n(a-k)^{m+1} + \frac{n}{m+1}I_{m+1,n-1}$ as req.

29) b)

$\int_0^1 \frac{x^4}{(2-x)^3}\,\mathrm{d}x$

$k = 1,\ a = 2,\ m = -3,\ n = 4$

$I_{m,n} = -\frac{1}{m+1} + \frac{n}{m+1}I_{m+1,n-1}$

$I_{-3,4} = \frac{1}{2} - 2I_{-2,3}$

$= \frac{1}{2} - 2\left(1 - 3I_{-1,2}\right) = -\frac{3}{2} + 6I_{-1,2}$

$= -\frac{3}{2} + 6I_{-1,2}$

Since $I_{-1,2} = -\frac{1}{0} + \frac{2}{0}I_{0,1}$ is indeterminate, now consider the integral:

$I_{-1,2} = \int_0^1 x^2(2-x)^{-1}\,\mathrm{d}x = \int_0^1 \frac{x^2}{2-x}\,\mathrm{d}x$

	$-x$	-2	
$-x$	x^2	$+2x$	$+4$
$+2$	$-2x$	-4	

$\frac{x^2}{2-x} = -x - 2 + \frac{4}{2-x}$

$\Rightarrow \int_0^1 \frac{x^2}{2-x}\,\mathrm{d}x = \int_0^1 \left(-x - 2 + \frac{4}{2-x}\right)\mathrm{d}x = \left[-\frac{1}{2}x^2 - 2x - 4\ln|2-x|\right]_0^1$

$= \left(-\frac{1}{2} - 2 - 0\right) - (-0 - 0 - 4\ln 2)$

$= 4\ln 2 - \frac{5}{2}$

So $I_{-3,4} = -\frac{3}{2} + 6\left(4\ln 2 - \frac{5}{2}\right) = 24\ln 2 - \frac{33}{2}$

$\therefore \int_0^1 \frac{x^4}{(2-x)^3}\,\mathrm{d}x = 24\ln 2 - \frac{33}{2}$

30) a)

$\lim_{x\to 0^+}(x\ln x) = \lim_{x\to 0^+}\left(\frac{\ln x}{\frac{1}{x}}\right) = \lim_{x\to 0^+}\left(\frac{\frac{1}{x}}{-\frac{1}{x^2}}\right) = \lim_{x\to 0^+}\left(\frac{x}{-1}\right) = 0$

30) b)

$I_n = \int_0^1 (\ln x)^n \, \mathrm{d}x$

Use Integration by Parts:

$$u = (\ln x)^n \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{n}{x}(\ln x)^{n-1} \qquad v = x$$

$\therefore \int_0^1 (\ln x)^n \, \mathrm{d}x = \lim_{a \to 0^+} [x(\ln x)^n]_a^1 - \int_0^1 n(\ln x)^{n-1} \, \mathrm{d}x$

From part a), we have $\lim_{x \to 0^+} (x \ln x) = 0$

For $n \geq 1$, we write $x(\ln x)^n = x \ln x \, (\ln x)^{n-1}$

As $x \to 0^+$, we have $\lim_{x \to 0^+} (x \ln x) = 0$ and we know that any power of $\ln x$ such as $(\ln x)^{n-1}$ grows very slowly so does not affect $x \ln x \to 0$

It follows that $\lim_{a \to 0^+} [a(\ln a)^n] = 0$

$\therefore \int_0^1 (\ln x)^n \, \mathrm{d}x = \lim_{a \to 0^+} [x(\ln x)^n]_a^1 - \int_0^1 n(\ln x)^{n-1} \, \mathrm{d}x$

$= \lim_{a \to 0^+} (1(\ln 1)^n - a(\ln a)^n) - n \int_0^1 (\ln x)^{n-1} \, \mathrm{d}x$

$= 0 - 0 - n \int_0^1 (\ln x)^{n-1} \, \mathrm{d}x$

$\therefore I_n = -nI_{n-1}$

30) c)

Prove true for $n = 0$:

$I_0 = \int_0^1 (\ln x)^0 \, \mathrm{d}x = \int_0^1 1 \, \mathrm{d}x = [x]_0^1 = 1$

$(-1)^n n! = (-1)^0 0!$=1 LHS is RHS $\therefore$ true for $n = 1$

Assume true for $n = k$:

$I_k = (-1)^k k!$

Prove true for $n = k + 1$:

$I_{k+1} = -(k+1)I_k = -(k+1)((-1)^k k!)$

$= -(-1)^k (k+1)!$

$= (-1)^{k+1}(k+1)!$

which is $I_n = (-1)^n n!$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

30) d)

$\int_0^1 (\ln x)^4 \, \mathrm{d}x = (-1)^4 4! = 24$